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Inflammation in preterm and term labour and delivery

Roberto Romero [a,b,c,*], Jimmy Espinoza [a,d], Luís F. Gonçalves [a,d], Juan Pedro Kusanovic [a], Lara A. Friel [d], Jyh Kae Nien [a]

[a] *Perinatology Research Branch, NICHD, NIH, DHHS, Bethesda, MD, USA*
[b] *Perinatology Research Branch, NICHD, NIH, DHHS, Detroit, MI, USA*
[c] *Center for Molecular Medicine and Genetics, Wayne State University, Detroit, MI, USA*
[d] *Department of Obstetrics and Gynecology, Wayne State University, Detroit, MI, USA*

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Innate immunity;
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Summary Inflammation has been implicated in the mechanisms responsible for preterm and term parturition, as well as fetal injury. Out of all of the suspected causes of preterm labour and delivery, infection and/or inflammation is the only pathological process for which both a firm causal link with preterm birth has been established and a molecular pathophysiology defined. Inflammation has also been implicated in the mechanism of spontaneous parturition at term.

Most cases of histopathological inflammation and histological chorioamnionitis, both in preterm and term labour, are sub-clinical in nature. The isolation of bacteria in the amniotic fluid, known as microbial invasion of the amniotic cavity, is a pathological finding; the frequency of which is dependent upon the clinical presentation and gestational age. There is a window of time during which it may be possible to detect a 'molecular signature of inflammation' by analysis of the transcriptome before histological evidence is observed.

This article reviews the role of inflammation in preterm and term parturition. It is possible that modulation of inflammation using anti-inflammatory cytokines, corticoids, antioxidants and/or other factors may complement antibiotic therapy and limit fetal injury.
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Introduction

Inflammation has been implicated in the mechanisms responsible for preterm and term parturition, as well as fetal injury. Out of all of the suspected causes of preterm labour (PTL) and delivery, infection and/or inflammation is the only pathological process for which both a firm causal link with preterm birth has been established and a molecular

* Corresponding author. Perinatology Research Branch, NICHD, NIH, DHHS, Hutzel Women's Hospital — Box #4, 3990 John R., Detroit, MI 48201, USA. Tel.: +1 313 993 2700; fax: +1 313 993 2694.
E-mail address: warfiela@mail.nih.gov (R. Romero).

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pathophysiology defined.[1] Inflammation has also been implicated in the mechanism of spontaneous parturition at term. This article reviews the role of inflammation in preterm and term parturition.

The spectrum of inflammation: clinical, histopathological and molecular definitions

Clinical inflammation has been classically defined by heat, pain, redness, swelling and impaired function, which reflect the effects of inflammatory mediators on local blood vessels and tissues. By contrast, histological inflammation, the gold standard for diagnosis, is defined by the infiltration of tissue by neutrophils, macrophages and lymphocytes. Chemotactic signals must be present for white blood cells (WBC) to migrate to the site of injury/infection. Thus, there is a window of time in which it may be possible to detect a 'molecular signature of inflammation' by analysis of the transcriptome (see below)[2] before histological evidence is observed.

There is a common misconception that the absence of systemic clinical signs, such as fever, leukocytosis, etc., makes inflammation/infection unlikely. In reality, the evidence suggests that the opposite holds true. Most cases of histopathological inflammation and chorioamnionitis, both in preterm[3,4] and term labour (Yoon and Romero, unpublished observations), are sub-clinical.

Inflammation accomplishes three main goals: (1) the delivery of cells and molecules for suppressing infection, (2) the generation of a physical barrier to the spread of infection, and (3) the promotion of repairs to the injured tissue.

Cells recruited to the site of injury release molecules during the course of inflammation, including antimicrobial peptides, cytokines and other inflammatory mediators. Some of these molecules alter macrophage and neutrophil activation so that microbial killing is enhanced. A second major goal of inflammation is to prevent the spread of microorganisms and this is often accomplished by the activation of the coagulation system and the formation of thrombi in blood vessels draining the site.

Injury can result from exposure to microorganisms or non-microbial-related insults. The immune system has evolved to identify non-self using pattern recognition receptors (PRR). These identify patterns of molecular structure common to most microorganisms. It is now known that PRR can also be used to identify 'danger signals.' The basic premise of the 'danger model' is that the immune system is more concerned with damage than foreignness ('non-self'), and that an immune reaction is initiated by 'alarm signals' from injured tissues, rather than by recognition of non-self.

Inflammation is part of the innate immune response, which acts immediately, is non-specific and lacks memory. Inflammation maintains tissue homeostasis. However, exaggerated inflammation or the lack of a response can lead to disease.

Infection as a cause of premature labour

Evidence of causality

Infection is a frequent and important mechanism of disease in PTL and delivery.[5,6] The evidence in support of this includes: (1) intrauterine infection or systemic administration of microbial products to pregnant animals can result in PTL and delivery,[7–10] (2) extrauterine maternal infections, such as pyelonephritis, have been associated with premature parturition, (3) sub-clinical intrauterine infections have been associated with PTL and delivery,[11] (4) patients with intra-amniotic infection or inflammation (defined as an elevation of the amniotic fluid (AF) concentration of cytokines)[12] in the midtrimester are at risk for subsequent preterm delivery, (5) antibiotic treatment of ascending intrauterine infections can prevent prematurity in experimental models of chorioamnionitis,[13] and (6) treatment of asymptomatic bacteriuria prevents prematurity.

Frequency and clinical significance of intrauterine infection

Intrauterine infections caused by bacteria are considered to be the leading cause of infection-associated preterm birth. Less than 1% of the AF of women not in labour at term will contain bacteria. Therefore, isolation of bacteria in the AF is a pathological finding, defined as microbial invasion of the amniotic cavity (MIAC). Most of these infections are sub-clinical and are not detected without AF analysis. The frequency of MIAC depends upon the clinical presentation and gestational age. MIAC occurs in 9% of asymptomatic patients with a short cervix (<25 mm sonographically) in the midtrimester.[14] In patients with PTL with intact membranes, the rate of positive (+) cultures is 12.8%.[6] However, among patients who have PTL with intact membranes and deliver preterm, the frequency is 22%. Among women with preterm premature rupture of membranes (PROM), the rate of +AF cultures at admission is 32.4%.[6] However, at the time of the onset of labour, as many as 75% of patients will have MIAC,[15] suggesting that microbial invasion occurs during the latent period. An interesting and consistent observation is that the lower the gestational age at presentation of PTL, the higher the frequency of +AF cultures.[16]

Microbiology of intrauterine infection

The most common microorganisms found in the amniotic cavity are genital *Mycoplasma* species, in particular, *Ureaplasma urealyticum*. However, other microorganisms found include *Mycoplasma hominis*, *Streptococcus agalactiae*, *Escherichia coli*, *Fusobacterium* species and *Gardnerella vaginalis*.

Significance of MIAC detected by molecular microbiology techniques

The prevalence of MIAC is based on the results of standard microbiological methods (i.e. cultivation), which are dependent upon culture conditions and on whether the growth requirements of the microorganisms are known. Of note, only 1% of the microbial world can be detected by cultivation ('the great plate count anomaly'). Consequently, the frequency of MIAC reported represents minimum estimates. In fact, several investigators have demonstrated that the prevalence of MIAC is higher when molecular techniques are used to detect conserved sequences in prokaryotes.[17,18]

Patients with a positive polymerase chain reaction (+PCR) for *U. urealyticum* but negative culture have adverse outcomes similar to patients with a +AF culture and they have worse outcomes than patients with sterile AF and a negative PCR.[19,20] Moreover, patients with a +PCR but a negative culture have the same degree of inflammation (AF interleukin (IL)-6 concentration, histological chorioamnionitis/funisitis) as those with a +AF culture.[20] Collectively, this evidence suggests that the presence of microbial footprints detected by PCR is associated with adverse outcomes.

Microorganisms in the chorioamniotic membranes: always pathological?

The amniotic cavity is normally considered to be sterile for bacteria, even when using molecular techniques. By contrast, fluorescent in situ hybridization (FISH) with a DNA probe specific for bacterial DNA has detected bacteria in the membranes of up to 70% of women undergoing a term elective caesarean delivery.[21] Bacteria are often present in the membranes of patients with PTL and intact membranes, as well as in patients with preterm PROM.[21] These findings suggest that the presence of bacteria alone is not sufficient to cause PTL and delivery and that microbial colonisation of the chorioamniotic membranes may not always elicit a fetal or maternal inflammatory response.

MIAC as a chronic process

Although chorioamnionitis is traditionally considered to be an acute process, evidence that MIAC exists for an extended period is mounting. Multiple observations have suggested that microbial invasion could be clinically silent in the midtrimester and that preterm delivery could take weeks to occur.[22]

Pathways of intra-amniotic infection

Microorganisms may gain access to the amniotic cavity and fetus using any of the following pathways: (1) ascending from the vagina/cervix (most commonly), (2) haematogenous dissemination through the placenta, (3) retrograde seeding from the peritoneal cavity, and (4) accidental introduction during invasive procedures (Fig. 1).[23]

There is some evidence to support a relationship between periodontal disease and PTL and delivery.[24] The mechanism underlying this association has not been definitively established. However, microorganisms found in the gingival crevice can be isolated from AF, suggesting that maternal bacteraemia and transplacental passage could account for some of these infections (see below).

Microbial products in the amniotic cavity

The adverse events associated with microbial invasion can be due to proliferation of intact microorganisms or of bacterial products. For example, the cell walls of Gram-negative bacteria contain lipopolysaccharide (LPS) or endotoxin, those of Gram-positive bacteria contain peptidoglycans (PGN) and lipoteichoic acid, while those of *Mycoplasma* contain

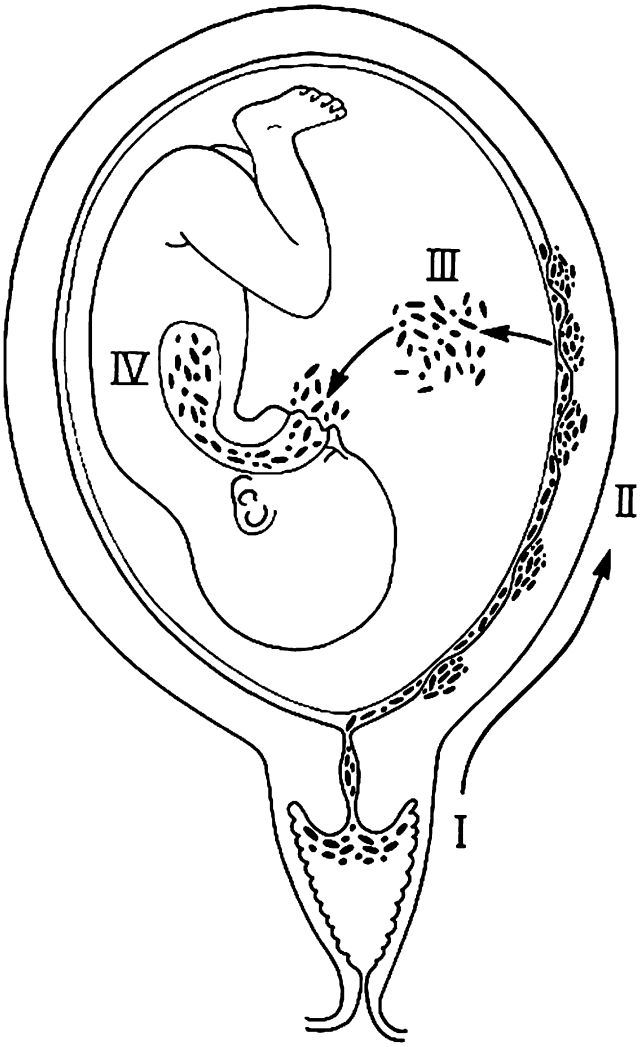

Figure 1 The ascending route of intrauterine infection. (I) Vaginal infection, (II) inflammation of fetal membranes, (III) microbial invasion of the amniotic cavity, and (IV) fetal inflammation.[24]

lipoglycans. Many of the effects of microorganisms are mediated by products that are released during bacterial death. Consequently, non-viable bacteria may exert deleterious effects if their products are recognised by PRR and elicit an inflammatory response. Indeed, concentrations of bacterial endotoxin in AF were found to be significantly higher in women with preterm PROM in labour than in women with PROM who were not in labour (Fig. 2).[25]

Detection of viral antigen in AF and adverse pregnancy outcome

Viral antigen is detected in the AF of up to 15% of asymptomatic low-risk pregnancies at the time of genetic amniocentesis[26] and in 41% of pregnancies with suspected viral infection based on ultrasound abnormalities.[27] The most common viral DNA isolates are adenovirus, cytomegalovirus (CMV) and enterovirus. Burguete et al.[28] determined that the prevalence of adeno-associated parvovirus by PCR analysis in midtrimester AF was 27%. Moreover, it has been found

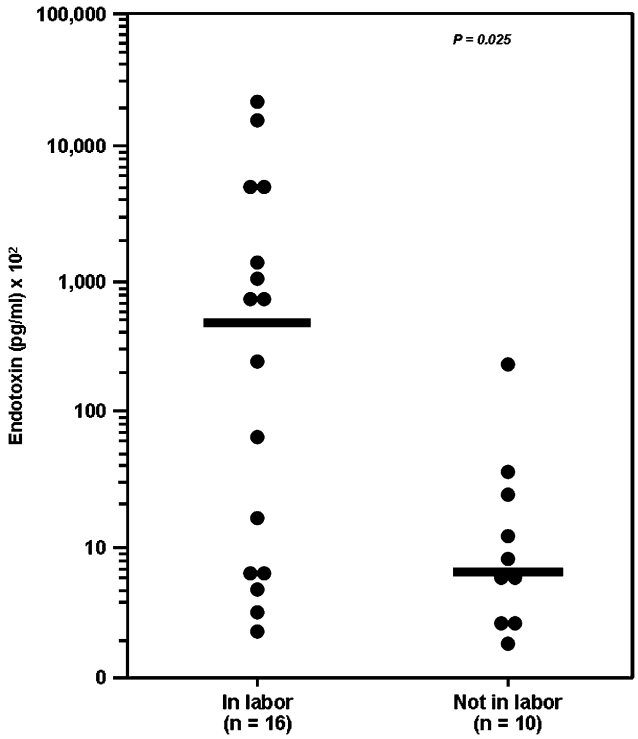

Figure 2 The concentration of microbial products in women with preterm premature rupture of membranes (PROM) in labour versus not in labour. Concentrations of microbial products were significantly higher in women with preterm PROM in labour.[25]

that among structurally normal fetuses, those with a +AF PCR for viral genome were more likely to die *in utero* and have both a lower gestational age and birth weight.

Inflammation as a mechanism for preterm parturition

An overview of the inflammatory response

Multiple anti-infective mechanisms protect normal pregnancy. Evidence suggests that bacteria cross intact chorioamniotic membranes.[29] Epithelium, however, represents more than a physical barrier against microorganisms. Most epithelia produce natural anti-microbial peptides (e.g. defensins, surfactant proteins), which can kill bacteria by damaging their cell membrane or facilitating phagocytosis.

Another mechanism of host defence is derived from the metabolic products of bacteria. For example, lactobacilli, which colonise the vagina shortly after birth, produce lactic acid and antimicrobial products.

The innate component of the immune system also provides immediate protection from microbial challenge by recognising the presence of microorganisms, thus preventing tissue invasion and/or eliciting a host response to limit proliferation/inflammation. One mechanism by which innate immunity recognises microorganisms is by using PRRs. PRRs, which are classified on the basis of function and sub-cellular localisation, include: (1) soluble PRRs, such as the 'acute phase proteins' mannan-binding lectin and C-reactive protein (CRP), which act as opsonins to neutralise and clear pathogens, (2) transmembrane PRRs, which include scavenger receptors, C-type lectins and Toll-like receptors (TLRs), and (3) intracellular PRRs, including Nod1 and Nod2, which mediate recognition of intracellular pathogens. Ten different TLRs, which recognise various ligands such as LPS, PGN, double-stranded RNA and flagellin, have been recognised in humans.

TLR-1, -2, -3, -5 and -6, have been identified in the epithelia from the vagina, ecto- and endocervix, endometrium and uterine tubes. TLR-4 has been demonstrated in the endocervix, endometrium and uterine tubes, but not in the vagina and ectocervix. Similarly, trophoblast cells are able to recognise and respond to pathogens through expression of TLRs, which may directly promote trophoblast cell death[30] observed in pregnancy complications.

The importance of TLRs in preterm parturition

Mice with a spontaneous mutation for TLR-4 are less likely to deliver preterm after intrauterine inoculation of heat killed bacteria or LPS than wild-type.[31] Moreover, spontaneous labour at term and preterm delivery with histological chorioamnionitis, regardless of membrane status, is associated with an increased mRNA expression of TLR-2 and TLR-4 in membranes.[32] These observations suggest that the innate immune system plays a role in parturition.

The role of pro-inflammatory cytokines (IL-1, TNF-α)

A solid body of evidence indicates that cytokines play a central role in the mechanisms of inflammation/infection-induced preterm parturition.[11,33] Evidence in support of the participation of IL-1 includes: (1) production by human decidua in response to bacterial products, (2) increased concentration and bioactivity in the AF of women with PTL and infection,[34] (3) stimulation of myometrial contractions (C. Bulletti, personal communication, 2002), and (4) induction of PTL and delivery by administration to pregnant animals, which was blocked by the administration of an antagonist.[35]

Similarly, evidence supporting the role of tumour necrosis factor-alpha (TNF-α) in mechanisms of preterm parturition includes: (1) TNF-α stimulates prostaglandin production by the amnion, decidua and myometrium, (2) human decidua can produce TNF-α in response to bacterial products,[36] (3) AF TNF-α bioactivity and immunoreactive concentrations are elevated in women with PTL and intra-amniotic infection,[36] (4) in women with preterm PROM and intra-amniotic infection, TNF-α concentrations are higher in the presence of labour,[37] (5) TNF-α application in the cervix induces changes that resemble cervical ripening,[38] and (6) TNF-α can induce preterm parturition when administered systemically to pregnant animals.

Redundancy in the cytokine network

Additional cytokines (interleukins, colony stimulating factors, etc.) and chemokines have also been implicated in the mechanisms of PTL and delivery. With the redundancy of the cytokine network implicated in parturition, the blockade of a single factor is insufficient to prevent preterm delivery in the context of infection.[39]

Anti-inflammatory cytokines and PTL

IL-10 is thought to be a key cytokine for the maintenance of pregnancy. Indeed, IL-10 production is significantly reduced in placentas of pregnancies at term not in labour compared with those from first- and second-trimesters, suggesting that downregulation of IL-10 is a physiological event favouring an inflammatory state temporal to the onset of labour.[40] IL-10 has also been implicated in the control of preterm parturition associated with inflammation. Finally, the administration of IL-10 in animal models of infection has been associated with improved pregnancy outcome.[41]

Thrombin, inflammation and PTL and delivery

Thrombin, a serine protease, plays a central role in blood coagulation by catalyzing conversion of fibrinogen into fibrin. This enzyme has also been implicated in platelet aggregation, endothelial cell and neutrophil activation, monocyte chemotaxis, lymphocyte proliferation and smooth muscle proliferation/contractility. With these multiple functions, thrombin has been proposed to play a central role in the regulation of inflammation and coagulation.

Accumulating evidence supports the role of thrombin in the mechanisms responsible for preterm parturition. Evidence in support of this includes: (1) decidua is a source of tissue factor, the primary initiator of coagulation, (2) intrauterine administration of whole blood to pregnant rats stimulates myometrial contractility in a dose-dependent manner, while heparinised blood does not,[42] (3) thrombin/anti-thrombin (TAT) complexes, a marker of thrombin generation, are increased in the plasma[43] and AF[44] of patients with PTL and preterm PROM, (4) an elevation of plasma TAT complex concentration in the second trimester is associated with subsequent preterm PROM,[45] and (5) the presence of retroplacental haematoma detected by first trimester ultrasound or vaginal bleeding in the first or second trimester is associated with adverse pregnancy outcomes.[46,47] The mechanisms responsible for increased generation of thrombin in PTL have not been determined.

The fetal inflammatory response syndrome

While the traditional definition of inflammation describes a local phenomenon, it is now recognised that inflammation may be present in circulating blood. Such a state, referred to as the 'systemic inflammatory response syndrome' (SIRS), is characterised by fever, tachycardia, hyperventilation and an elevated WBC count. These symptoms have been attributed to the effects of pro-inflammatory mediators. In 2001 certain mediators, such as IL-6, were found to be associated with SIRS, an observation that may bring about a new, more specific, definition of SIRS. We defined the fetal counterpart of SIRS, the 'fetal inflammatory response syndrome' (FIRS), in 1997, using the same parameters as those proposed for adults: an elevated fetal blood IL-6 concentration of IL-6 $>$ 11 pg/ml.[48,49]

FIRS was originally described in pregnancies complicated by PTL and preterm PROM. Fetuses with FIRS had a higher rate of severe neonatal morbidity (e.g. respiratory distress syndrome, suspected/proven neonatal sepsis, pneumonia, bronchopulmonary dysplasia, intraventricular haemorrhage, periventricular leukomalacia, or necrotising enterocolitis)[49] and a shorter cordocentesis-to-delivery interval.

Many findings have since been confirmed using umbilical cord blood at birth, including the relationship between pro-inflammatory cytokines and the likelihood of clinical and suspected sepsis. Pathological examination of the umbilical cord is an alternative approach to determine whether fetal inflammation was present before birth. Funisitis and chorionic vasculitis are the histopathological hallmarks of FIRS.[50] Funisitis is associated with endothelial activation, a key mechanism in the development of organ damage. Neonates with funisitis are at increased risk for neonatal sepsis[4] and long-term handicaps, such as bronchopulmonary dysplasia and cerebral palsy.[51] In addition, since neutrophils in the AF are predominantly of fetal origin, the AF WBC count can be used as an indirect index of fetal inflammation.[52] Intra-amniotic inflammation is a risk factor for impending preterm delivery and adverse perinatal outcome in women with preterm PROM, even in the absence of documented intra-amniotic infection.[53]

Fetal microbial invasion or other insults resulting in FIRS can progress toward multiple organ dysfunction, shock and perhaps death in the absence of timely delivery. Several fetal organs, including the haematopoietic system, adrenals, heart, brain, lungs and skin, have been proposed to be target organs during FIRS (Fig. 3; our unpublished results).

The role of the fetus in the onset of PTL

Among women with preterm PROM, FIRS is associated with the onset of PTL, regardless of the AF inflammatory state (Fig. 4).[54] This suggests that the human fetus plays a role in initiating labour; however, maternal cooperation must occur. Thus, it is possible that systemic fetal inflammation may occur in the absence of labour when the inflammatory process does not involve the chorioamniotic membranes and decidua, particularly in the context of haematogenous infections or other disease processes (i.e. alloimmunisation).

Fetal death and maternal/fetal systemic inflammation

The rate of maternal inflammation is nine times more frequent than that of fetal inflammation in stillbirth. This could have two potential explanations. First, it is possible that fetal infection occurs, but fails to trigger a fetal inflammatory response and labour. In this case, *in utero* fetal death would represent failure of the host response to infection. Secondly, inflammation of placental membranes may occur after fetal death and be unrelated to the demise.

Infection in term labour

Microbial invasion of the amniotic cavity may be detected in approximately 18% of patients in labour at term[55] and in 30% of patients with PROM at term.[56] Although the microbial isolates are similar to those observed in PTL, no difference has been observed in the rates of chorioamnionitis, cervical dilatation and neonatal outcome between culture-positive and culture-negative patients. The frequency

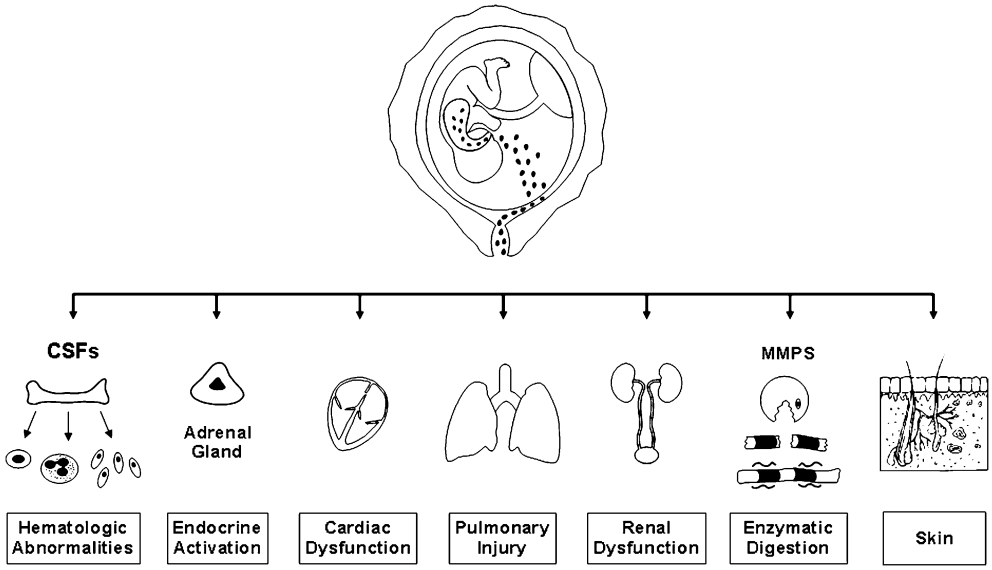

Figure 3 Fetal organs proposed to be target organs during the fetal inflammatory response syndrome (FIRS) are the haematopoietic system, adrenals, heart, brain, lungs and skin (our unpublished results). CSFs, colony stimulating factors; MMPs, matrix metalloproteinases.

of fetal invasion is extremely low, as is the intensity of the cytokine response to microorganisms and their products in the amniotic cavity.[55]

Inflammation in term labour

Parturition as an inflammatory process

Liggins was the first to liken cervical ripening to an inflammatory response. Since then, accumulating evidence supports the idea that inflammation can be detected in the cervix, myometrium, chorioamniotic membranes and amniotic cavity of women in labour. Spontaneous labour at term is associated with the infiltration of inflammatory cells in these tissues and increased production of pro-inflammatory cytokines (IL-1β, IL-6, TNF-α and IL-8)[33,34] and chemokines (GROα, granulocyte colony stimulating factor (GCSF), GMCSM, neutrophil attractant/activating peptide-1/IL-8, etc).[33] Recently, using a genome-wide screen, we have been able to demonstrate that genes involved in the control of inflammation are upregulated in the chorioamniotic membranes after labour at term, even in the absence of histological chorioamnionitis. Of interest is that an

		n	Procedure-to-Delivery Interval (median, range, days)
I	AF IL-6 $\leq$ 7.9 ng/ml FP IL-6 $\leq$ 11 pg/ml	14	5 (0.2-33.6)
II	AF IL-6 > 7.9 ng/ml FP IL-6 $\leq$ 11 pg/ml	5	7 (1.5-32)
III	AF IL-6 > 7.9 ng/ml FP IL-6 > 11 pg/ml	6	1.2 (0.25-2)
IV	AF IL-6 $\leq$ 7.9 ng/ml FP IL-6 > 11 pg/ml	5	0.75 (0.13-10)

Figure 4 The association of the fetal inflammatory response syndrome (FIRS) and preterm labour. Among women with preterm premature rupture of membranes (PROM), FIRS, as measured by fetal serum (FP) concentration of interleukin (IL)-6 $>$ 11 pg/ml, is associated with the impending onset of preterm labour, regardless of the inflammatory state of the amniotic fluid (AF).[54]

inflammatory signature was not observed in the maternal circulation of patients with spontaneous labour at term.[2] This suggests that an inflammatory process is localised to the membranes. These findings may reflect the fact that spontaneous parturition is associated with an increased pro-inflammatory cytokine and chemokine response.[33]

FIRS at term

In the context of MIAC, a fetal cytokine response, as assessed by IL-6 concentrations in umbilical cord plasma, has been demonstrated to be more intense in preterm than term neonates. This has been observed even among neonates born to mothers with a negative AF culture.[57] There are two possible explanations for this: (1) the fetal attack rate by microorganisms in the AF is higher in preterm than term gestation and/or (2) there are differences in the cytokine response to microbial products in these fetuses. Congenital sepsis is more frequent in the preterm than the term neonate and this may be related to an increased susceptibility to infection by the preterm fetus. The other possibility is that MIAC leading to PTL and preterm PROM may be chronic in nature and prolonged exposure to microorganisms increases the likelihood of fetal infection and response. Indeed, there is evidence that patients with a positive AF culture for microorganisms at the time of mid-trimester genetic amniocentesis are at increased risk for prematurity and pregnancy loss,[23] as are patients with evidence of intra-amniotic inflammation.

Recent observations indicate that funisitis can occur in the absence of clinical chorioamnionitis (B.H. Yoon et al., unpublished results). In a study of 832 consecutive patients who delivered term singleton neonates within 72 h of amniocentesis, the prevalence of funisitis was 4% (30/832). Most patients with funisitis (83% (25/30)) also had histological chorioamnionitis, suggesting that when funisitis occurs, a maternal inflammatory response has been deployed. Of note, 17% of fetuses had funisitis without histological chorioamnionitis. Potential explanations for this include a systemic transplacental infection that affects the fetus, but not the membranes. Alternatively, a sampling bias may occur if chorioamnionitis does not involve the entire membrane. The first explanation is unlikely, because villitis would be expected in cases of isolated funisitis, which was not observed. In addition, although funisitis was associated with MIAC, histological chorioamnionitis and intra-amniotic inflammation, none of the patients with funisitis ($n = 30$) had clinical chorioamnionitis. Thus, funisitis at term appears to have a different clinical significance than preterm funisitis.

A role of the fetus in the onset of labour

Recently, it has been demonstrated that the fetus plays a role in the initiation of mouse parturition. Evidence supporting this view includes: (1) intra-amniotic administration of SP-A to mice resulted in labour, (2) treatment with anti-SP-A antibody prolonged gestation, (3) the concentration of SP-A in AF increased with advancing gestation from 17 days postcoitum until term, and (4) AF macrophages migrate to the uterus with increasing AF SP-A. The authors proposed that production of SP-A by the fetal lung near term causes activation/migration of fetal macrophages to the maternal uterus, where increased IL-1β leads to labour.[58]

The maternal systemic inflammatory response in normal pregnancy and PTL and delivery

Normal pregnancy has been proposed to involve physiological activation of the innate limb of the immune response. Preterm parturition with intact or ruptured membranes is associated with phenotypic and metabolic changes in maternal monocytes and granulocytes that are consistent with the presence of intravascular maternal inflammation.[59,60]

Conclusions

The evidence reviewed indicates that parturition at term is an inflammatory process. Intra-amniotic infection/inflammation is causally linked to preterm parturition, the development of FIRS in a subset of patients, and fetal injury. Moreover, preterm parturition (with intact or ruptured membranes) is associated with a maternal systemic inflammatory response characterised by phenotypic and metabolic changes of maternal monocytes and granulocytes. It is possible that modulation of inflammation with anti-inflammatory cytokines, corticoids, antioxidants and/or other factors may complement antibiotic therapy and limit fetal injury.

Acknowledgement

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Practice points

- Of all suspected causes of preterm labour and delivery, infection and/or inflammation is the only pathological process for which both a firm causal link with preterm birth has been established and a molecular pathophysiology defined.
- Most cases of histopathological inflammation and histological chorioamnionitis, both in preterm and term labour, are sub-clinical in nature.
- The immune system has evolved to identify the non-self using pattern recognition receptors, e.g. Toll-like receptors, which identify patterns of molecular structure common to most microorganisms.
- The isolation of bacteria in the amniotic fluid, microbial invasion of the amniotic cavity (MIAC), is a pathological finding, the frequency of which is dependent upon the clinical presentation and gestational age.
- The presence of microbial footprints detected by PCR is associated with adverse outcomes.
- The most common pathway that microorganisms use to gain access to the amniotic cavity and fetus is ascending from the vagina/cervix.

- Epithelia represent more than a physical barrier against microorganisms by the production of natural antimicrobial peptides (e.g. defensins and surfactant proteins), which can kill bacteria by damaging their cell membrane or facilitating phagocytosis.
- A solid body of evidence indicates that cytokines, including interleukin (IL)-1, play a central role in the mechanisms of inflammation/infection-induced preterm parturition.
- Thrombin has been proposed to play a role in preterm parturition through the regulation of both inflammation and coagulation.
- Funisitis and chorionic vasculitis are the histopathologic hallmarks of the fetal inflammatory response syndrome (FIRS).
- Funisitis is associated with endothelial cell activation, a key mechanism in the development of organ damage.
- Neonates with funisitis are at increased risk for neonatal sepsis and long-term handicap, such as bronchopulmonary dysplasia and cerebral palsy.
- MIAC may be detected in approximately 18% of patients in labour at term and in 30% of patients with premature rupture of membranes (PROM) at term. Although the microbial isolates are similar to those observed in preterm labour (PTL), no difference has been observed in the rates of chorioamnionitis, cervical dilatation and neonatal outcome between culture-positive and culture-negative patients.
- The frequency of fetal invasion at term in patients with MIAC is extremely low, as is the intensity of the cytokine response to microorganisms and their products in the amniotic cavity.

Research directions

- Use of a molecular signature for prediction/risk assessment of those patients destined to have preterm labour (PTL) and preterm premature rupture of membranes (PROM).
- Enhance detection of sub-clinical infection/inflammation, possibly using transcriptomics/proteomics, since early treatment may improve neonatal/maternal outcomes.
- Manipulate innate immune system to improve surveillance/suppress infection, keeping balance with its protective role.
- Explore the mechanisms by which epithelia actively inhibit infection and mechanisms to enhance their action/potential.
- Manipulate cytokine balance to promote successful pregnancy.
- Elucidate the fetal role in initiation and perpetuation of preterm and term labour and delivery.

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